



Name: Ayush Chaubey

PRN: 25070122058

Division: A2

ASSIGNMENT No: 4

TITLE: Develop a Java application demonstrating the use of wrapper classes for conversion between primitive data types and objects, and performing basic operations on wrapped values.

Problem Statement: Develop a Java application demonstrating the use of wrapper classes for conversion between primitive data types and objects, and performing basic operations on wrapped values.

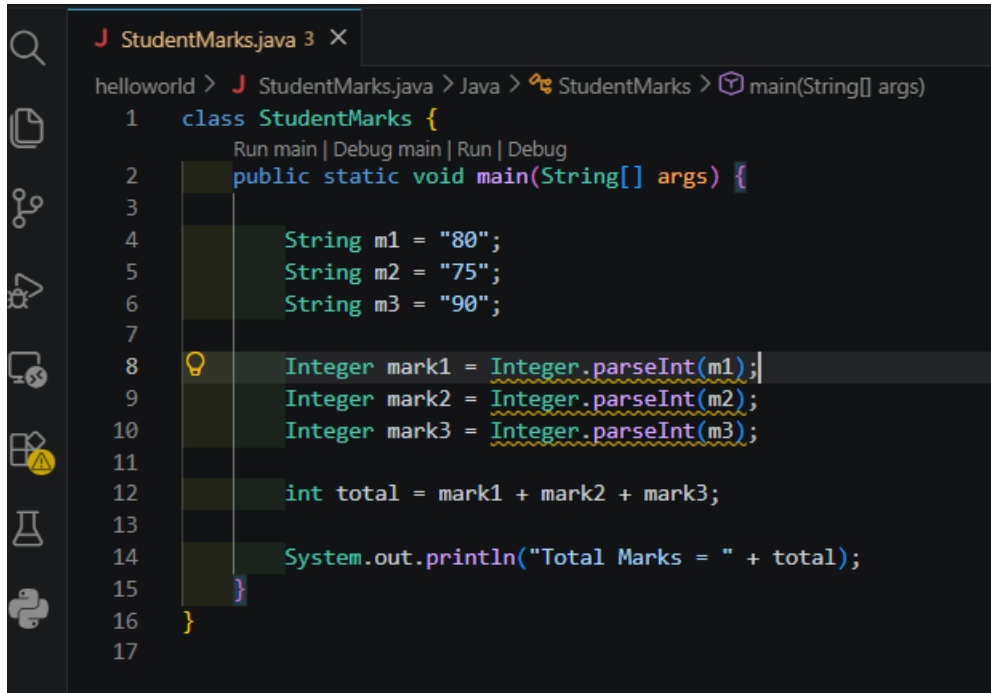
Learning Objectives: To perform conversion between primitive data types and wrapper objects using Java wrapper classes.

Theory: Wrapper classes are predefined classes that encapsulate primitive data types into objects. Java provides wrapper classes such as Integer, Double, Character, and Boolean for every primitive type. They support autoboxing, unboxing, parsing, and conversion methods. Wrapper classes are widely used in Java Collections Framework and when object representation of primitive values is required.

Exercise:

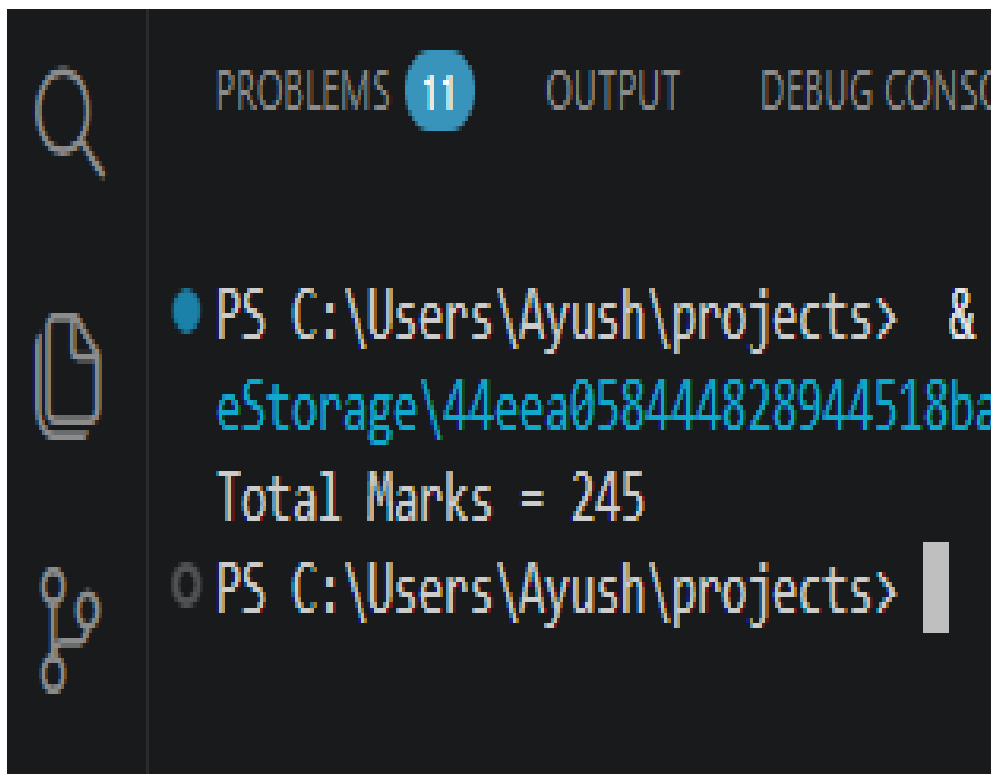
1. Create a program to convert student marks from String format to Integer and calculate total marks.

Program:



```
1 class StudentMarks {  
2     public static void main(String[] args) {  
3  
4         String m1 = "80";  
5         String m2 = "75";  
6         String m3 = "90";  
7  
8         Integer mark1 = Integer.parseInt(m1);  
9         Integer mark2 = Integer.parseInt(m2);  
10        Integer mark3 = Integer.parseInt(m3);  
11  
12        int total = mark1 + mark2 + mark3;  
13  
14        System.out.println("Total Marks = " + total);  
15    }  
16 }  
17
```

Output:



```
PROBLEMS 11 OUTPUT DEBUG CONSOLE  
● PS C:\Users\Ayush\projects> &  
eStorage\44eea058444828944518ba  
Total Marks = 245  
○ PS C:\Users\Ayush\projects>
```

2. Develop an Employee Payroll System that accepts employee IDs, basic salary, and bonus amounts from the user. Convert the entered values into wrapper objects and perform validation operations to ensure valid salary values before calculating the net salary.

Program:

```
EmployeePayroll.java 1 X
helloworld > J EmployeePayroll.java > ...
1  import java.util.Scanner;
2
3  class EmployeePayroll {
4      public static void main(String[] args) {
5
6          Scanner in = new Scanner(System.in);
7
8          System.out.print(s: "Enter Employee ID: ");
9          String id = in.next();
10
11         System.out.print(s: "Enter Basic Salary: ");
12         String basic = in.next();
13
14         System.out.print(s: "Enter Bonus: ");
15         String bonus = in.next();
16
17         Integer empId = Integer.valueOf(id);
18         Double basicSalary = Double.valueOf(basic);
19         Double bonusAmount = Double.valueOf(bonus);
20
21         if (basicSalary >= 0 && bonusAmount >= 0) {
22             double netSalary = basicSalary + bonusAmount;
23
24             System.out.println("Employee ID: " + empId);
25             System.out.println("Net Salary = " + netSalary);
26         } else {
27             System.out.println(x: "Invalid Salary or Bonus!");
28         }
29     }
30 }
```

Output:

```
File Edit Selection View Go Run Terminal
PROBLEMS 16 OUTPUT DEBUG CONSOLE TERMINAL
PS C:\Users\Ayush\projects> & 'C:\Users\Ayush\workspaceStorage\44eeea058444828944518bac730833f76\redl
● PS C:\Users\Ayush\projects> c:; cd 'c:\Users\
AppData\Roaming\Code\User\workspaceStorage\44e
Enter Employee ID: 121
Enter Basic Salary: 200000
Enter Bonus: 10000
Employee ID: 121
Net Salary = 210000.0
○ PS C:\Users\Ayush\projects> 
```